

Table 1: Comparing the performance of dominating (DG) and dominated (DD) groups under the numeric step neighborhood. [†] and * denote that the performance difference exhibits $p < 0.001$ and $0.001 \leq p < 0.05$, respectively. Green rows and red rows denote the cases that confirm and violate our theory of model usefulness, respectively.

System	No aggregation			Majority vote			Weighted sum		
	Δp	DG Lose (%)	DG Win (%)	Δp	DG Lose (%)	DG Win (%)	Δp	DG Lose (%)	DG Win (%)
APACHE	-0.0191	39.1	60.9	-0.0028	49.1	50.9	-0.0255	43.6	56.4
7z	-145.3881 [†]	22.7	77.3	-46.4100*	42.0	58.0	-167.5533 [†]	30.6	69.4
DCONVERT	-0.0016*	45.3	54.7	-0.0003	47.3	51.5	-0.0030*	41.1	57.8
DEEPARCH	-0.0079 [†]	18.0	82.0	-0.0011 [†]	29.1	67.4	-0.0011 [†]	30.8	65.8
EXASTENCILS	-192.1372 [†]	17.2	82.8	-23.7633*	42.2	57.8	-121.1267 [†]	25.8	74.2
HADOOP	-0.3222	47.7	52.3	-0.2667	47.5	52.5	-0.3333	45.0	55.0
MARIADB	-0.0779	39.1	60.9	-0.0453	45.4	54.6	-0.0436	45.6	54.4
MONGODB	-417.9015 [†]	25.8	74.2	-8.8733*	38.1	51.2	-27.8900 [†]	32.8	56.1
POSTGRESQL	2.3402*	59.4	40.6	-1.0633	47.0	53.0	5.7733 [†]	63.6	36.4
REDIS	-28.4553	45.3	54.7	23.8118	52.6	47.4	-119.5436	41.4	58.6
SPARK	-3.7204 [†]	32.0	68.0	-5.2500 [†]	37.8	61.9	-4.8667 [†]	38.6	61.1
STORM	-0.0000	24.2	63.3	0.0000	47.0	39.2	0.0000	38.3	47.2
HSMGP	0.8928 [†]	80.5	19.5	0.2686 [†]	59.6	39.2	0.0315	54.4	44.4
XGBOOST	-0.9543 [†]	25.0	75.0	-0.1782 [†]	39.6	59.8	-0.3178 [†]	35.6	63.9
HIPACC	0.0172 [†]	71.9	28.1	0.0162 [†]	64.4	35.6	0.0065*	56.4	43.6
SQLITE	-0.0173 [†]	34.4	65.6	-0.0179*	43.0	57.0	-0.0024*	48.1	51.9
JAVAGC	-1.3544 [†]	21.1	78.9	-0.6250*	39.2	60.8	-0.2917*	45.8	54.2
POLLY	-0.0836 [†]	14.8	85.2	-0.0212 [†]	31.7	68.3	-0.0411 [†]	27.2	72.8

Table 2: Comparing the performance of dominating (DG) and dominated (DD) groups under the adjacent bin neighborhood. [†] and * denote that the performance difference exhibits $p < 0.001$ and $0.001 \leq p < 0.05$, respectively. Green rows and red rows denote the cases that confirm and violate our theory of model usefulness, respectively.

System	No aggregation			Majority vote			Weighted sum		
	Δp	DG Lose (%)	DG Win (%)	Δp	DG Lose (%)	DG Win (%)	Δp	DG Lose (%)	DG Win (%)
APACHE	-0.0191	39.1	60.9	-0.0028	49.1	50.9	-0.0255	43.6	56.4
7z	-161.4285 [†]	21.9	78.1	-33.6333*	43.6	56.4	-175.2067 [†]	30.6	69.4
DCONVERT	-0.0016*	45.3	54.7	-0.0003	47.3	51.5	-0.0030*	41.1	57.8
DEEPARCH	-0.0079 [†]	18.0	82.0	-0.0011 [†]	29.1	67.4	-0.0011 [†]	30.8	65.8
EXASTENCILS	-192.1372 [†]	17.2	82.8	-23.7633*	42.2	57.8	-121.1267 [†]	25.8	74.2
HADOOP	-0.7329 [†]	42.2	57.8	-1.6833*	41.5	58.2	0.1333	50.6	49.2
MARIADB	-0.0779	39.1	60.9	-0.0453	45.4	54.6	-0.0436	45.6	54.4
MONGODB	-417.9015 [†]	25.8	74.2	-8.8733*	38.1	51.2	-27.8900 [†]	32.8	56.1
POSTGRESQL	2.3402*	59.4	40.6	-1.0633	47.0	53.0	5.7733 [†]	63.6	36.4
REDIS	-15.3580	48.4	51.6	21.8592	50.9	49.1	-93.2930	43.1	56.9
SPARK	-3.6475 [†]	32.0	68.0	-4.5500 [†]	40.9	58.8	-5.6167 [†]	37.2	62.5
STORM	-0.0000	24.2	63.3	0.0000	47.0	39.2	0.0000	38.3	47.2
HSMGP	0.8928 [†]	80.5	19.5	0.2686 [†]	59.6	39.2	0.0315	54.4	44.4
XGBOOST	-0.9543 [†]	25.0	75.0	-0.1782 [†]	39.6	59.8	-0.3178 [†]	35.6	63.9
HIPACC	0.0146 [†]	65.6	34.4	0.0129 [†]	60.9	39.1	0.0074*	57.5	42.5
SQLITE	-0.0173 [†]	34.4	65.6	-0.0179*	43.0	57.0	-0.0024*	48.1	51.9
JAVAGC	-1.4236 [†]	19.5	80.5	-0.6250*	39.2	60.8	-0.2917*	45.8	54.2
POLLY	-0.0836 [†]	14.8	85.2	-0.0212 [†]	31.7	68.3	-0.0411 [†]	27.2	72.8